國立虎尾科技大學
108-2期末會考解答

(一)填充題:

1. 設$x+y+z=105$,求$2xy+2yz+2zx$最大值=
   解:

   設目標函數為$f(x,y,z)=2xy+2yz+2zx$，限制函數為$g(x,y,z)=x+y+z-105$
   令$F(x,y,z,\lambda)=f(x,y,z)+\lambda g(x,y,z)=2xy+2yz+2zx+\lambda(x+y+z-105)$
   令$\begin{cases} F_x=2y+2z+\lambda=0 \\ F_y=2x+2z+\lambda=0 \\ F_z=2x+2y+\lambda=0 \\ F_\lambda=x+y+z-105=0 \end{cases}$　　由$F_x-F_y\Longrightarrow y=x$, $F_x-F_z\Longrightarrow z=x$
   即$x=y=z$
   代入$F_\lambda=0$得$x=35$, $y=35$, $z=35$, 故$(35,35,35)$為相對極點
   此外，任選滿足限制條件$g(x,y,z)=0$之(x,y,z)，例如$(x,y,z)=(50,50,5)$
   $\because f(50,50,5)=6000<f(35,35,35)=7350$, 故極大值為7350

2. 求$\displaystyle\int_1^4\int_0^x\sqrt{x-y}\ dydx=$
   解:

   $$\int_1^4\int_0^x\sqrt{x-y}\ dydx=\int_1^4 -\frac{2}{3}(x-y)^{\frac{3}{2}}\Big|_{y=0}^{x}dx=\frac{2}{3}\int_1^4 x^{\frac{3}{2}}\,dx$$
   $$=\frac{2}{3}\Big(\frac{2}{5}x^{\frac{5}{2}}\Big|_1^4\Big)=\frac{2}{3}\Big\{\frac{2}{5}(32-1)\Big\}$$
   $$=\frac{124}{15}$$

3. 設$f(x,y)=\cos(x+y)+xy^2$,求$f_x(\frac{-\pi}{2},0)=$
   解:

   $f_x=-\sin(x+y)+y^2$
   $f_x(\dfrac{-\pi}{2},0)=-\sin(\dfrac{-\pi}{2}+0)+0^2=1$

4. 設 $\Omega=\{(x,y)|1\le x^2+y^2\le 4\}$,求$\displaystyle\iint_\Omega e^{x^2+y^2}\ dxdy=$

解:

令$x=r\cos\theta$, $y=r\sin\theta$, $dxdy=rdrd\theta$

$$\iint\limits_{\Omega} e^{x^2+y^2}\,dxdy=\iint\limits_{\Omega} e^{r^2}rdrd\theta\,,\ \Omega=\{(r,\theta)|1\le r\le 2,0\le\theta\le 2\pi\}$$

$$=\int_0^{2\pi}\int_1^2 re^{r^2}drd\theta=\int_0^{2\pi}\frac{e^{r^2}}{2}\Big|_{r=1}^{2}d\theta=\int_0^{2\pi}\frac{1}{2}(e^4-e^1)d\theta=\frac{\theta}{2}(e^4-e^1)\Big|_0^{2\pi}=\pi(e^4-e^1)$$

5. 求 $\displaystyle\lim_{(x,y)\to(0,0)}\frac{3^y\tan^{-1}(2x)}{x}$ $=$

解:

$$\lim_{(x,y)\to(0,0)}\frac{3^y\tan^{-1}(2x)}{x}=\lim_{(x,y)\to(0,0)}3^y\cdot\lim_{(x,y)\to(0,0)}\frac{\tan^{-1}(2x)}{x}=1\cdot\lim_{(x,y)\to(0,0)}\frac{\frac{2}{1+4x^2}}{1}=2\cdot 1=2$$

6. 已知 $x^2-3y^2+2z\ =\sin(z-1)$,求$\displaystyle(\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y})\Big|_{(1,1,1)}$ $=$

解:

令$F(x,y,z)=x^2-3y^2+2z-\sin(z-1)$, 則$F_x=2x$, $F_y=-6y$, $F_z=2-\cos(z-1)$

$$\therefore\frac{\partial z}{\partial x}=-\frac{F_x}{F_z}=-\frac{2x}{2-\cos(z-1)}\ ,\ \frac{\partial z}{\partial y}=-\frac{F_y}{F_z}=-\frac{-6y}{2-\cos(z-1)}$$

$$\Rightarrow\left(\frac{\partial z}{\partial x}+\frac{\partial z}{\partial y}\right)\Big|_{(1,1,1)}=\frac{-2x+6y}{2-\cos(z-1)}\Big|_{(1,1,1)}=\frac{-2+6}{2-\cos(1-1)}=4$$

7. 設 $f(x,y)=\ln(x^2y^2+ex-ey+e)$, 求$f_y(0,0)$ $=$

解:

$$f_y=\frac{2x^2y-e}{x^2y^2+ex-ey+e}$$

$$f_y(0,0)=\frac{0-e}{0+0-0+e}=-1$$

8. 求 $\displaystyle\lim_{(x,y)\to(0,0)}\frac{x^2}{\sqrt{x^2+y^2}}$ $=$

解:

令$x = r\cos\theta$, $y = r\sin\theta$

則 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2}{\sqrt{x^2+y^2}} = \lim_{r\to 0}\frac{r^2\cos^2\theta}{r} = \lim_{r\to 0}(r\cos^2\theta)$

$\because 0 \le \cos^2\theta \le 1 \Rightarrow 0 \le r\cos^2\theta \le r$, 且 $\displaystyle\lim_{r\to 0} 0 = 0$, $\displaystyle\lim_{r\to 0} r = 0$

由夾擠定理可知 $\displaystyle\lim_{r\to 0} r\cos^2\theta = 0$，故 $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2}{\sqrt{x^2+y^2}} = 0$

9. 求 $\displaystyle\int_0^2\int_1^2 (3-y)x^2\ dydx =$

解:

$$\int_0^2\int_1^2 (3-y)x^2\ dydx = \int_0^2 (3y - \frac{y^2}{2}\Big|_1^2)x^2\,dx = \frac{3}{2}\int_0^2 x^2\,dx = \frac{3}{2}\Big(\frac{x^3}{3}\Big|_0^2\Big) = \frac{3}{2}\cdot\frac{8}{3} = 4$$

10. 已知函數$f(x,y)$滿足$f_x(1,-1) = 4$, $f_y(1,-1) = 3$,求$\displaystyle\frac{d}{dt}f(3t+1, \sin(4t)-1)\Big|_{t=0} =$

解:

令$z = f(3t+1, \sin(4t)-1)$,則$z = f(x,y)$, $x = 3t+1$, $y = \sin(4t)-1$

由連鎖法則得 $\displaystyle\frac{dz}{dt} = \frac{\partial f(x,y)}{\partial x}\times\frac{dx}{dt} + \frac{\partial f(x,y)}{\partial y}\times\frac{dy}{dt} = f_x(x,y)\times 3 + f_y(x,y)\times 4\cos 4t$

$\because t = 0 \Rightarrow x = 1$, $y = -1$

$$\frac{d}{dt}f(3t+1, \sin(4t)-1)|_{t=0} = f_x(1,-1)\times 3 + f_y(1,-1)\times 4\cos(4\times 0) = 4\times 3 + 3\times 4 = 24$$

11. 設 $f(x,y) = e^{-2x}\cos(2y)$,求$f_{xx}(x,y) - f_{yy}(x,y) =$

解:

$f_x = -2e^{-2x}\cos(2y)$, $f_{xx} = 4e^{-2x}\cos(2y)$

$f_y = -2e^{-2x}\sin(2y)$, $f_{yy} = -4e^{-2x}\cos(2y)$

$f_{xx}(x,y) - f_{yy}(x,y) = 4e^{-2x}\cos(2y) - (-4e^{-2x}\cos(2y)) = 8e^{-2x}\cos(2y)$

12. 設 $f(x,y) = xy - \dfrac{1}{x} - \dfrac{1}{y}$,求$f_{xy}(1,1) - f_{yx}(1,1) =$

解:

$f_x = y + \dfrac{1}{x^2}$, $f_{xy} = 1$

$f_y = x + \dfrac{1}{y^2}$, $f_{yx} = 1$

$f_{xy}(1,1) - f_{yx}(1,1) = 1 - 1 = 0$

13. 設 $\Omega=\{(x,y)|x^2+y^2\leq 1, y\geq 0\}$, 求 $\iint_\Omega (x^2+y^2)\ dxdy=$

解:

令 $x=r\cos\theta$, $y=r\sin\theta$, $dxdy=rdrd\theta$

$$\iint\limits_\Omega (x^2+y^2)\,dxdy=\iint\limits_\Omega r^2 rdrd\theta\,,\ \Omega=\{(r,\theta)|0\leq r\leq 1, 0\leq\theta\leq\pi\}$$

$$=\int_0^\pi\int_0^1 r^3 drd\theta=\int_0^\pi\left(\frac{r^4}{4}\Big|_0^1\right)d\theta=\int_0^\pi\frac{1}{4}d\theta=\frac{1}{4}\theta\Big|_0^\pi=\frac{\pi}{4}$$

14. 設 $f(x,y)=x^3y^2, x=u^2+2v, y=uv^2$, 求 $\frac{\partial f}{\partial u}\Big|_{(u,v)=(1,-1)}=$

解:

由連鎖律可知 $\dfrac{\partial f}{\partial u}=\dfrac{\partial f}{\partial x}\cdot\dfrac{\partial x}{\partial u}+\dfrac{\partial f}{\partial y}\cdot\dfrac{\partial y}{\partial u}$

$\dfrac{\partial f}{\partial x}=3x^2y^2$, $\dfrac{\partial f}{\partial y}=2x^3y$, $\dfrac{\partial x}{\partial u}=2u$, $\dfrac{\partial y}{\partial u}=v^2$

$\Longrightarrow \dfrac{\partial f}{\partial u}=3x^2y^2\cdot 2u+2x^3y\cdot v^2$

當 $(u,v)=(1,-1)\Rightarrow x=-1, y=1$

$\dfrac{\partial f}{\partial u}\Big|_{(x,y)=(-1,1)}^{(u,v)=(1,-1)}=3\cdot 1\cdot 2-2\cdot(-1)^2=4$

(二)計算題:

1. 求函數 $f(x,y)=3x^2+y^3+6x-6y-6xy$ 的臨界點、相對極值與鞍點。

解:

$f_x=6x+6-6y$, $f_y=3y^2-6-6x$

$f_{xx}=6$, $f_{xy}=-6$, $f_{yy}=6y$

$F(x,y)=f_{xx}\cdot f_{yy}-{f_{xy}}^2=36y-36$

令 $\begin{cases} f_x=6x+6-6y=0\\ f_y=3y^2-6-6x=0\end{cases}$

$\Longrightarrow f_x+f_y=3y^2-6y=0$, $3y(y-2)=0\Longrightarrow y=0,2$, 帶入 $f_x=0$ 可得 $x=-1,1$

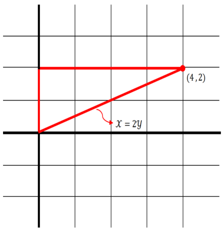
$\Rightarrow$ 得 $(-1,0),(1,2)$ 為臨界點

(x,y)	$F(x,y)$	$f_{xx}(x,y)$	$f(x,y)$
$(-1,0)$	<0	>0	鞍點
$(1,2)$	>0	>0	極小

$\therefore$ 極小值為 $f(1,2)=7$

2. 若Ω是以$(0,0),(0,2),(4,2)$為頂點之三角型區域， 求$\displaystyle\iint_\Omega e^{y^2}dxdy$

解:



$$\therefore \Omega=\{(x,y)|0\le x\le 2y, 0\le y\le 2\}$$

$$\iint\limits_\Omega e^{y^2}\,dxdy=\int_0^2\int_0^{2y}e^{y^2}\,dxdy$$

$$=\int_0^2 xe^{y^2}\Big|_{x=0}^{2y}dy=\int_0^2 2ye^{y^2}\,dy=e^{y^2}\Big|_0^2=e^4-1$$

3. 試求多變數極限 $\displaystyle\lim_{(x,y)\to(0,0)}\frac{x^2-y^2}{x^2+y^2}$

解:

若(x,y)沿x軸$(y=0)$趨近(0,0)，則$\displaystyle\lim_{x\to 0}\frac{x^2-0}{x^2+0}=\lim_{x\to 0}\frac{x^2}{x^2}=\lim_{x\to 0}1=1$

若(x,y)沿y軸$(x=0)$趨近(0,0)，則$\displaystyle\lim_{y\to 0}\frac{0-y^2}{0+y^2}=\lim_{y\to 0}\frac{-y^2}{y^2}=\lim_{y\to 0}-1=-1$

$\because 1\neq -1 \therefore \displaystyle\lim_{(x,y)\to(0,0)}\frac{x^2-y^2}{x^2+y^2}$不存在